

HIGH-FREQUENCY PLASMA-CHEMICAL SYNTHESIS AND EXTRACTION OF PRECIOUS METALS FROM REFRACTORY SULFIDE ORES OF THE KYZYLKUM REGION

Author: Usmanov R.
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ABSTRACT

This paper presents the results of an experimental study on non-contact high-frequency electromagnetic exposure applied to a refractory gold-bearing mineral—arsenopyrite (FeAsS) from the Kokpatas and Daugyztau deposits (Central Kyzylkum) in the presence of a deuterated environment. It was established that under conditions of sharp resonance of the high-frequency electromagnetic field (HF EMF) and non-equilibrium deuterium plasma, disintegration of the mineral's crystal lattice occurs, followed by the deposition of elemental gold onto the receiving antenna. The presence of the precious metal on the copper collector was confirmed via fire assay at the Central Research Laboratory (CNIL). Utilizing gamma-activation analysis (Central Laboratory of GAA, Zarafshan) on a linear electron accelerator, an anomalous isotopic composition of the synthesized gold was detected, including the isotopes Au-196, Au-197, Au-198, and Au-199. A hypothesis regarding the endogenous nucleosynthesis of precious metals within the Earth's crust under identical physicochemical and wave-induced conditions is proposed.

1. Introduction and Problem Statement

The processing of refractory gold-arsenic ores of the Central Kyzylkum region via traditional metallurgical methods (bio-oxidation, autoclave oxidation) involves substantial energy expenditures and severe environmental risks. Gold within arsenopyrites (FeAsS) resides predominantly in a finely dispersed ('invisible') isomorphic state locked inside the crystal lattice.

The objective of this study was to evaluate a fundamentally new physicochemical method for unlocking the sulfide matrix and extracting encapsulated metals through targeted, resonant high-frequency exposure within an oxygen-free, isotope-modified medium.

2. Experimental Methodology and Materials

Samples of refractory arsenopyrite (a total of 27 distinct ore samples investigated) obtained from the Kokpatas and Daugyztau deposits served as the object of study.

Experimental Setup Architecture:

1. **Reaction Zone:** The mineral sample was positioned inside a hermetically sealed, oxygen-free cell directly above a vessel containing the liquid phase.
2. **Isotopic Medium Modifier:** The liquid phase consisted of a heavy water (deuterium) condensate, produced via kinetic isotope separation during the controlled solar-driven evaporation of a natural water body conducted over a 2–3 day period.
3. **Field Source and Emitter:** Electromagnetic exposure was delivered via a specialized copper tubular HF-range antenna coupled to a high-frequency generator.

Experimental Protocol: Upon applying power to the HF EMF system, the oscillating circuit was brought into a state of sharp wave resonance. Due to an intense skin effect and a high surface current density, the temperature of the copper antenna exceeded the melting point of copper (1085°C). Simultaneously, rapid evaporation of the underlying deuterated liquid took place. Upon reaching the incandescent surface of the antenna, the deuterium vapors underwent ionization, generating a dense torch of non-equilibrium high-frequency plasma acting as a process catalyst.

3. Results and Discussion

3.1. Macroscopic Transformations and Fire Assay

Under the influence of the deuterium HF-plasma, both thermal and electrodynamic destruction of the arsenopyrite crystal structure occurred. Ions of the liberated (or newly synthesized) metal were transported by electromagnetic forces toward the maximum field gradient zone, depositing onto the surface of the melting copper tube. A distinct 'gilding' effect was macroscopically observed on the surface of the copper antenna due to deep thermal diffusion. The melted elements of the antenna were transferred to the CNIL facility. Standard fire assay (cupellation method) unequivocally verified the presence of elemental gold within the copper alloy, registering a tangible material yield of the precious metal on the collector. Concurrently, the entire series of the 27 HF EMF-irradiated raw arsenopyrite ore samples was prepared for subsequent nuclear-physical verification.

3.2. Nuclear-Physical Verification and Isotopic Spectrum

To eliminate matrix interference and meticulously investigate the nuclear structure of the recovered metal, the melted elements of the copper antenna along with all 27 HF EMF-irradiated ore samples were dispatched to the Central Laboratory of Gamma-Activation Analysis (CL GAA) in Zarafshan. All submitted specimens and samples were subjected to bombardment by an intense flux of high-energy bremsstrahlung gamma radiation generated by a linear electron accelerator striking a refractory (tungsten) target. Measurements using a semiconductor gamma spectrometer were conducted under rigorous timing protocols. The automated spectrometry complex recorded a distinct peak at 409 keV on the computer monitor, corresponding to the Au-197m isomer (half-life of 7.7 s), which validates the presence of the stable nuclide Au-197. However, beyond the stable phase, the spectrometer identified an anomalous isotopic chain of gold: Au-196, Au-198, and Au-199, as well as correlating lines for silver (Ag) and uranium (U).

4. Physical Interpretation and Geophysical Conclusions

The detection of the short-lived isotopes Au-198 and Au-199 demonstrates that during the experiment, the gold nuclei were exposed to a dense flux of free neutrons. The dynamics behind the formation of this isotopic spectrum spanning masses 196 to 199 yield proof of low-energy nuclear processes (transmutation or isotopic exchange) taking place directly within the non-equilibrium deuterium HF-plasma at the 'copper-arsenopyrite' phase boundary.

Geophysical Hypothesis: The empirical data gathered from this laboratory simulation can be extrapolated to interpret large-scale endogenous processes within the Earth's lithosphere. The subterranean architecture of the Kyzylkum deposits functions effectively as natural wave reactors. Tectonic shifts along deep-seated fault zones generate localized ultra-high pressures, seismogenic currents, and piezoelectric fields capable of instantaneously heating micro-volumes of the formation above 1000°C, driving semiconducting sulfides (FeAsS) into a plasma state. In the presence of deep-seated fluids naturally enriched with deuterium, a continuous process of endogenous nucleosynthesis ('birth') of gold and associated elements is triggered within the pores of the mineral matrix. This necessitates a fundamental revision of classical cosmogonic models governing the formation of ore deposits.